SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I Nighil Adhithyan *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	7
Experimental Design	30	8
Use of Tools	20	1
Analysis of Results	15	2
Documentation	10	3

Total Marks 21 **/30**

Signature of the Faculty

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TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

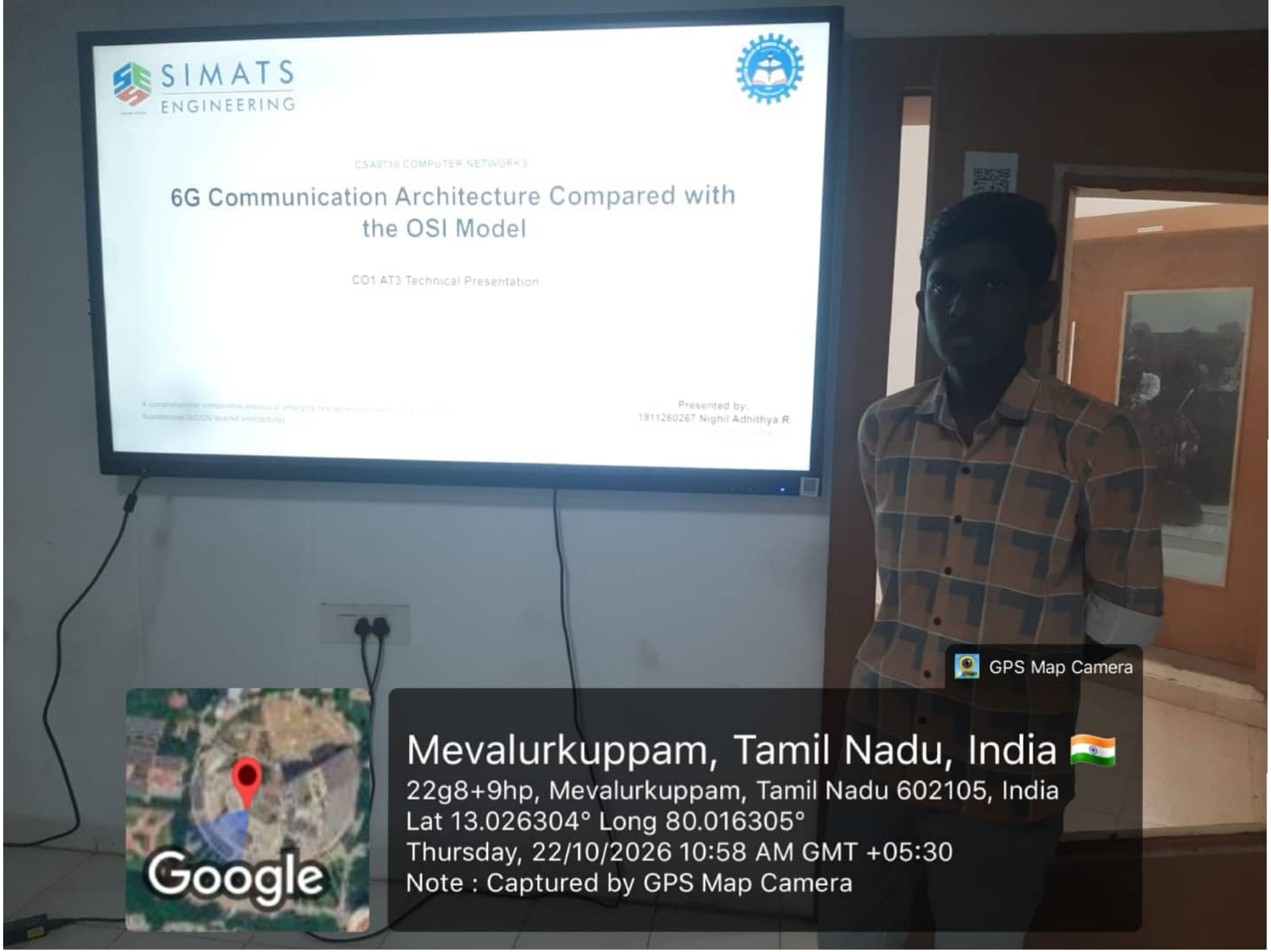
RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

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			interpretation		
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

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SIMATS ENGINEERING

Engineer to Excel

CSA0710 COMPUTER NETWORKS

6G Communication Architecture Compared with the OSI Model

CO1 AT3 Technical Presentation

Presented by:
Nighil Adhithya Ramesh (1911260267)

Department of Computer Science

A comprehensive comparative analysis of emerging next-generation wireless paradigms against foundational ISO/OSI layered architectures

Foundational Context & Course Alignment

Connecting core Computer Networks course objectives (CO1) with advanced telecommunication trends

OSI Baseline Functions

- Structured 7-layer hierarchy ensuring modular interoperability.
- Reliable data encapsulation and peer-to-peer communication.
- Standardized addressing, error control, and flow management.

Course Outcome CO1 Focus

6G Paradigm Shift

- Transition from reactive layers to proactive AI control planes.
- Terahertz wireless spectrum enabling unprecedented speeds.
- Convergence of communication, sensing, and computing.

Next-Gen Infrastructure

Research Problem

- Rigid ISO/OSI boundaries introduce serialization bottlenecks.
- Ultra-low sub-millisecond latency targets require flattening.
- Evaluation of cross-layer optimization vs. strict encapsulation.

Critical Evaluation

The Traditional ISO/OSI Reference Model Architecture

Detailed breakdown of the 7 hierarchical layers governing structured data communication

Layers 7–5	**Application, Presentation, Session Layers** Handling user interfaces, syntax translation, data compression, and secure session state control.	HTTP / TLS / DNS
Layer 4	**Transport Layer** End-to-end reliable data delivery, flow control, windowing mechanisms, and congestion management.	TCP / UDP / QUIC
Layer 3	**Network Layer** Logical packet addressing, subnet routing, path selection algorithms, and ICMP diagnostics.	IPv4 / IPv6 / OSPF
Layers 2–1	**Data Link & Physical Layers** Node-to-node frame synchronization, MAC addressing, error detection, and raw bit stream transmission.	Ethernet / Wi-Fi

Evolutionary Leap: From 5G to 6G Networks

Examining core performance indicators and technological drivers leading toward 6G systems

Terahertz (THz) Spectrum

- Frequencies ranging between 0.1 THz and 10 THz.
- Massive bandwidth supporting over 1 Tbps peak rates.
- Requires advanced directional beamforming arrays.

Peak Data Rate: > 1 Tbps

Sub-Millisecond Latency

- End-to-end delays reduced below 0.1 milliseconds.
- Essential for real-time haptic feedback and holography.
- Eliminates jitter in autonomous industrial systems.

Latency Target: < 0.1 ms

AI-Native Architecture

- Machine learning embedded intrinsically in protocols.
- Autonomous self-optimization of radio resources.
- Predictive routing based on user mobility patterns.

Control Plane: Distributed ML

6G Architectural Vision & Core Pillars

Transforming telecommunication infrastructure into intelligent, programmable distributed fabrics

Distributed Cloud & Edge

- Compute nodes embedded directly within base stations.
- Decentralized execution of latency-critical tasks.
- Reduces backhaul congestion to core data centers.

Edge Intelligence

Reconfigurable Surfaces

- Smart metamaterials embedded on building facades.
- Actively shapes radio wave reflection and refraction.
- Overcomes high propagation loss in THz bands.

Smart Environment

Joint Comm & Sensing

- Simultaneous wireless communication and radar imaging.
- Enables precise environmental mapping and tracking.
- Optimizes spectrum utilization through shared waveforms.

JCAS Framework

OSI L1–L2 vs. 6G: Physical & Data Link Layers

Comparative evaluation of transmission media, frame control, and error handling mechanisms

Traditional OSI Paradigm

- **Fixed Framing:** Strict MAC boundaries and static frame sizes optimized for Ethernet/Wi-Fi topologies.
- **Passive Media:** Copper and fiber optic cables carrying modulated RF signals with fixed attenuation models.
- **Layer Isolation:** ARQ and FEC operating independently per layer without cross-layer state feedback.

Baseline: Encapsulated Framing

6G Architecture Paradigm

- **RIS Metamaterials:** Programmable walls actively shaping electromagnetic wave propagation paths in real time.
- **THz Air Interfaces:** Ultra-high frequency wireless links requiring molecular absorption and beam tracking.
- **AI-Driven MAC:** Dynamic channel access adapting instantly to traffic bursts using predictive estimators.

Advanced: Smart Metamaterials

OSI L3–L4 vs. 6G: Network & Transport Layers

Transitioning from IP packet routing and TCP congestion control to semantic routing and AI transport

Traditional OSI Paradigm

- **IP Addressing:** Packet forwarding based on hierarchical IPv4/IPv6 headers and longest-prefix matching.
- **TCP/UDP Control:** End-to-end congestion management, window scaling, and reliable byte streams.
- **Rigid Multiplexing:** Strict port-based separation of data flows and independent socket abstractions.

Baseline: IP & TCP/UDP

6G Architecture Paradigm

- **Semantic Routing:** Transmitting data meaning and feature maps rather than raw redundant packet bits.
- **Predictive Transport:** Machine learning anticipates network congestion before packet loss or queue drops occur.
- **In-Network Computing:** Processing and filtering data directly inside routing nodes to reduce core overhead.

Advanced: Semantic & Predictive

OSI L5–L7 vs. 6G: Upper Layers & Applications

From client-server sessions to decentralized AI services and immersive holographic applications

Traditional OSI Paradigm

- **Client-Server Model:** Centralized cloud APIs, HTTP/DNS request-response flows, and static database queries.
- **Session Handshakes:** Explicit TLS negotiation and socket establishment protocols for secure transport.
- **Syntactic Formatting:** JSON/XML structured payload serialization across standardized protocols.

Baseline: Client-Server APIs

6G Architecture Paradigm

- **Distributed AI Fabrics:** Swarm intelligence, collaborative edge agents, and federated machine learning apps.
- **Immersive XR & Holograms:** Multi-sensory real-time spatial data streams requiring ultra-low jitter.
- **Zero-Touch Orchestration:** Autonomous self-healing service management and intent-driven networking.

Advanced: Decentralized AI

Why 6G Challenges the Traditional OSI Stack

Vertical encapsulation overhead and isolated layer design create significant bottlenecks in ultra-low latency environments

Encapsulation Latency

- Sequential traversal across 7 distinct protocol layers.
- Header serialization bloat accumulates processing delay.
- Impractical for sub-millisecond tactile internet needs.

Processing Overhead

Cross-Layer Synergy

- Physical layer radio metrics must inform app codecs.
- Strict isolation prevents rapid adaptation to fading.
- Direct state sharing optimizes buffer allocations.

Direct State Sharing

AI-Centric Flattening

- Global neural networks replace rigid layer rules.
- End-to-end data delivery path optimization.
- Bypasses traditional encapsulation barriers safely.

Neural Optimization

Summary & Future Research Directions

Key takeaways connecting computer networks fundamentals with future telecommunications architecture

OSI Legacy Endures

- Fundamental networking concepts remain conceptually vital.
- Addressing, framing, and structured reliability provide base logic.
- Essential pedagogical foundation for engineers.

Conceptual Foundation

Protocol Convergence

- Rigid static layering replaced by dynamic AI pipelines.
- Software-defined networking principles extended to wireless.
- Real-time adaptive packet and stream handling.

Dynamic Pipelines

Academic Impact

- Fulfills Computer Networks course outcomes successfully.
- Explores advanced networking trends and protocol performance.
- Bridges classroom theory with cutting-edge 6G research.

Course Outcome CO1

Thank You